

4.8.2 Red-shift (physics only)

Content	Key opportunities for skills development
There is an observed increase in the wavelength of light from most distant galaxies. The further away the galaxies, the faster they are moving and the bigger the observed increase in wavelength. This effect is called red-shift. The observed red-shift provides evidence that space itself (the universe) is expanding and supports the Big Bang theory.	
The Big Bang theory suggests that the universe began from a very small region that was extremely hot and dense. Since 1998 onwards, observations of supernovae suggest that distant galaxies are receding ever faster. Students should be able to explain: • qualitatively the red-shift of light from galaxies that are receding • that the change of each galaxy's speed with distance is evidence of an expanding universe • how red-shift provides evidence for the Big Bang model • how scientists are able to use observations to arrive at theories such as the Big Bang theory	WS 1.2
• that there is still much about the universe that is not understood, for example dark mass and dark energy.	WS 1.1, 1.3